

Optimization of mRNA Secondary Structure Prediction Using Quantum Computing

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Moderna Challenge

Live demo harold-ohandja.github.io/Moderna-Quantum-RNA-/

Shape decides the medicine

- + An RNA strand folds back on itself into a two-dimensional shape.
- + That shape drives stability, how efficiently the mRNA makes protein, and manufacturability.
- + Predicting the shape well is part of designing a better medicine.
- + The difficulty: the number of possible folds explodes as the strand gets longer.



The problem, precisely

Input

A sequence over A, U, C and G.

Target

The Minimum Free Energy (MFE) folding.

Complexity

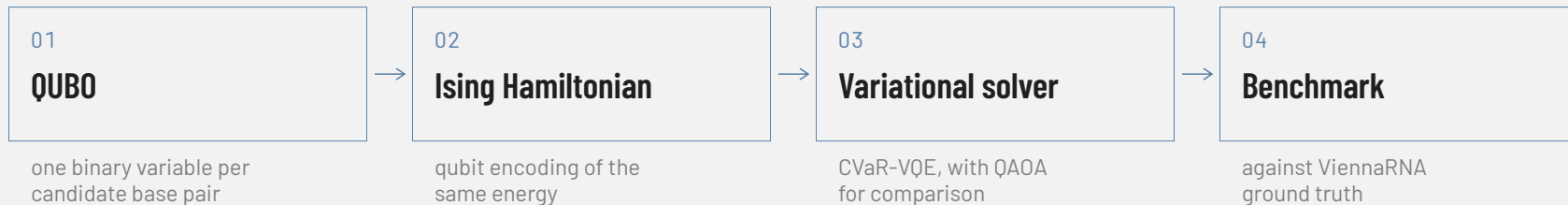
Classical tools solve the nested case in $O(L^3)$. Adding pseudoknots makes it NP-hard.

Our goal, per the challenge: reproduce known structures for small sequences and study how quantum resources scale — not beat classical methods.

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A hybrid quantum-classical pipeline



The pipeline mirrors the method published by Moderna and IBM Quantum (arXiv:2405.20328).



The QUBO formulation

$$H = E_{\text{thermodynamic}} + P_{\text{overlap}} + P_{\text{pseudoknot}}$$

Energy

-1.0 base pair · -1.5 stacking

Rewards a pair, and rewards stacked pairs more.

Overlap

+5.0 penalty

No base may take part in two pairs at once.

Pseudoknot

+5.0 penalty

Crossing pairs are excluded, so solutions stay nested.

One binary variable per candidate base pair — 1 means paired. Weights in kcal/mol.

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The quantum solvers

Primary — CVaR-VQE

$\alpha = 0.1$ on a hardware-efficient n_{local} ansatz. It optimises the best 10% of sampled bitstrings, which stays robust in rugged energy landscapes.

Comparison — QAOA

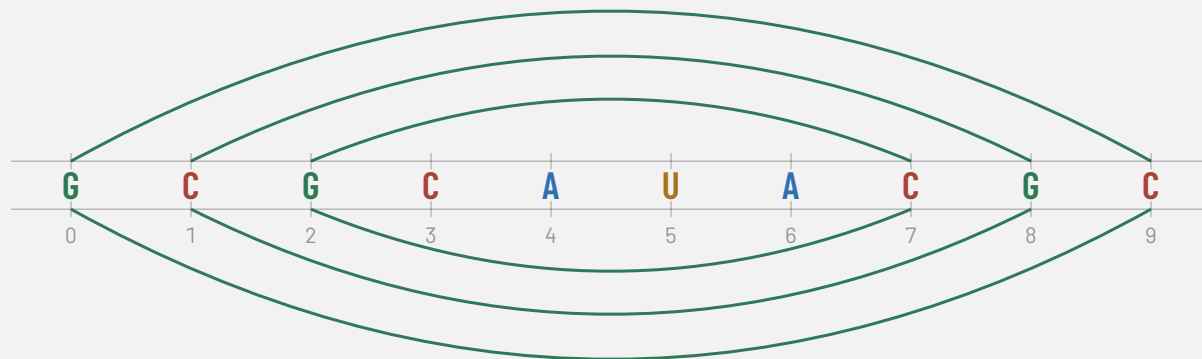
QAOA and CVaR-QAOA solve the same Hamiltonian, so the result is attributable to the formulation rather than one optimiser.

All runs on Qiskit Aer — simulation, no hardware required under the challenge rules.



Exact match on a 10-nt sequence

CVAR-VQE · (((.....)))



VIENNARNA GROUND TRUTH · (((.....)))

■ A ■ U ■ C ■ G

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0.0

kcal/mol energy gap to the
classical MFE

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solvers reproduce the
exact structure



Where it misses, and why

Every method misses

On AUGCAUGC and
GCAUCGUAGC, no solver
returns the reference structure.

Brute force too

Exhaustive search over the
same QUBO misses it as well —
it cannot beat the objective it
is given.

So: a formulation gap

The simplified energy model
rewards a pairing real
thermodynamics would not.
The optimiser is not at fault.

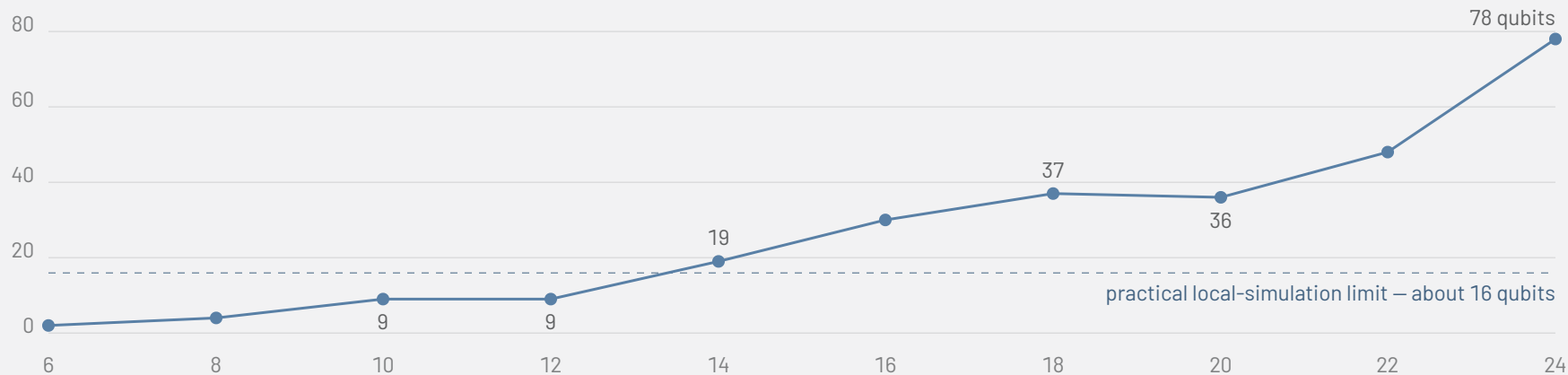
Separating a bad model from a weak optimiser is the point of a benchmark. We document the failures rather than report a clean sweep.





Qubits required by length

one qubit per candidate base pair



RNA sequence length (nucleotides) against required qubits / binary variables. QUBO terms grow with the same candidate-pair count, so circuit width and classical simulation cost rise together.

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Runs recovering the exact MFE structure – gap 0.0 kcal/mol – across finite-shot sampling noise and hardware-inspired depolarizing gate noise on the smallest sequence. The caveat is honest: a shallow circuit on one small instance, and no basis for extrapolating to long sequences.





Bonus tasks and future work

Completed

Pseudoknot exclusion, with justification. Noise robustness study.

Partial

Comparison of multiple QUBO encodings.

Next

A richer nearest-neighbour energy model to close the formulation gap; a pseudoknot-aware formulation; real-hardware runs; longer sequences.



Reproducible

CVaR-VQE mirroring the
sponsor's published method.

Honest benchmarking,
failures included.

Live demo: harold-ohandja.github.io/Moderna-Quantum-RNA-/
Repository and notebooks linked from the demo page.